

A Quantitative Photochemical Experiment for Undergraduates

The aim of this experiment is to introduce chemistry students to the concept of the quantum yield of a chemical reaction, i.e., the relationship between the rate at which a particular process occurs from an excited electronic state of a molecule to the rate at which the molecule absorbs radiation. Generally the product of a photochemical reaction in solution must be separated from the starting material, solvent, and any other products before quantitative estimation can be carried out. Even if the techniques for performing such separations were available in the teaching laboratory, the process would often be too lengthy compared with the time available in a practical period. The present experiment was set up to overcome this difficulty, since no separation of the product from the starting material is necessary for its quantitative spectrophotometric estimation. The experiment has advantages in that the ferrioxalate actinometer used to measure the light input is also dependent upon the use of a spectrophotometer, the apparatus used is simple and fairly readily obtained, and the whole experiment can be conveniently carried out in a single 3-hr laboratory period.

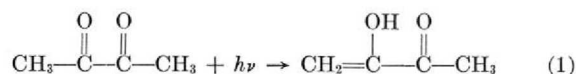
¹ LEMAIRE, J., *J. Phys. Chem.*, **71**, 2653 (1967).

² HATCHARD, C. G., AND PARKER, C. A., *Proc. Roy. Soc. (London)*, **A235**, 518 (1956).

³ CALVERT, J. G., AND PITTS, JR., J. N., "Photochemistry," John Wiley & Sons, New York, p. 783.

Basis of the Experiment

It has been shown that biacetyl, when irradiated at around 300 nm in deaerated aqueous solution undergoes a photochemical enolization¹



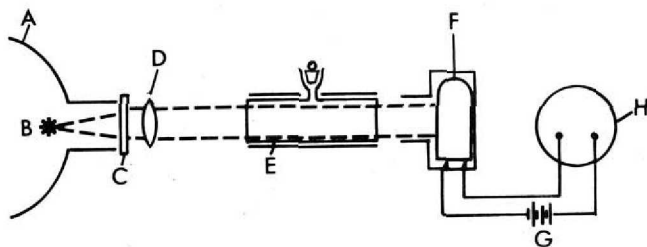
Since the end form is conjugated, it displays a strong new ultraviolet absorption band with a maximum centered at 275 nm. Thus a measure of the optical density at this wavelength of an irradiated solution of biacetyl relative to an unirradiated solution will give a quantitative measure of the amount of enol formed per unit time during the irradiation provided the molar extinction coefficient of the enol is known. This may be taken¹ as 5000 mole⁻¹ cm⁻¹.

In order to calculate the quantum yield of enolization, it is necessary to know the rate in quanta per second at which light is absorbed. The students can be told that an absolute determination using a thermopile is an exacting and difficult procedure, and thus a chemical actinometer is used. The most convenient and frequently used is the ferrioxalate actinometer due to Parker.² Here a standard procedure is used in which ferrous ions formed by irradiation of the ferrioxalate are complexed with *o*-phenanthroline, and quantitatively measured on a spectrophotometer at 510 nm. Know-

ing the molar extinction coefficient of the Fe(II)-*o*-phenanthroline complex ($1.11 \times 10^4 \text{ l mole}^{-1} \text{ cm}^{-1}$)³ and the quantum yield of Fe²⁺ formation in the particular wavelength region used (1.23), the intensity of radiation emitted by the lamp can be determined in quanta/sec. For details see reference in footnote 3.

Experimental

Light from a Hanovia medium pressure mercury arc (Fluorescence Mode 11) fitted with a light shield was focussed by a Pyrex lens into an approximately parallel beam which passed through a Corning 9863 filter into a cylindrical Pyrex reaction vessel, 7 cm long, 3.5 cm diameter. The volume of the vessel was ~78 ml, and the outside was painted black to cut out stray light. After traversing the reaction cell, the light was metered using an R.C.A. 935 phototube. The output from this phototube was monitored on a spot galvanometer, (see figure). The spread of wavelengths of light passing



Apparatus for photochemical experiment. A, metal light shield; B, Hanovia medium pressure mercury arc; C, Corning 9863 filter; D, lens; E, Pyrex reaction cell in metal sleeve and holder; F, light shielded RCA 935 phototube; G, 90 V radio battery; H, spot galvanometer.

through the cell was thus between 3000 and 4000 Å, since the Pyrex cut off at approximately 3000 Å and the filter at 4000 Å.

For the actinometry, a 0.006 *M* solution of potassium ferrioxalate solution was introduced into the cell and irradiated for approximately 12 min (the particular experimental set-up used this time was found to give a convenient optical density for the complex solution). A fraction (10 ml) of the irradiated solution was pipetted into a 25-ml volumetric flask, and 5 ml sodium acetate buffer solution and 2 ml *o*-phenanthroline (0.1%) solution were added. The solution was then made up to the mark, mixed well, and allowed to stand in the dark for 30 min before the optical density at 510 nm was measured on a Hilger and Watts SP500 spectrophotometer in a 1-cm cell.

The total number of Fe²⁺ ions, n_{Fe} , produced in the reaction is given by the equation

$$n_{Fe} = \frac{6.023 \times 10^{20} V_1 V_3 \log_{10} I_0/I}{V_2 I \epsilon_i}$$

where V_1 is the volume of actinometer solution irradiated; V_2 , the volume of aliquot for analysis (10 ml); V_3 , the final volume to which aliquot is diluted (25 ml); $\log_{10} I_0/I$, the optical density of solution relative to un-

irradiated solution at 510 nm; 1, the path length of spectrophotometer cell (1 cm); and ϵ_i the molar extinction coefficient of Fe²⁺ phenanthroline complex ($= 1.11 \times 10^4 \text{ mole}^{-1} \text{ l cm}^{-1}$).

If the quantum yield of Fe²⁺ ion formation is taken as 1.23, and knowing the fraction of incident light absorbed in the cell by the actinometer solution (should be 100%) and the time of irradiation, the intensity of radiation emitted by the lamp in quanta/sec can be calculated.

For the enolization, a $1.8 \times 10^{-2} M$ solution of biacetyl in water was degassed by bubbling with oxygen-free nitrogen for 30 min (while the actinometry was carried out). This solution was then introduced quickly into the reaction vessel, firmly stoppered, and irradiated for 120 min. A fraction was then removed and the optical density relative to an unirradiated solution at 275 nm was measured on the same spectrophotometer as before. The number of enol molecules formed during the radiation, n_{enol} , is given by

$$n_{enol} = \frac{6.023 \times 10^{20} V_1 \log_{10} I_0/I_{enol}}{\epsilon_2 l}$$

where V_1 and 1 are as before, $\log_{10} I_0/I$ is the optical density of irradiated biacetyl solution, and ϵ_2 is the molar extinction coefficient of enol ($= 5000 \text{ mole}^{-1} \text{ l cm}^{-1}$). The quantum yield of enol formation Q_{enol} , is then

$$Q_{enol} = \frac{n_{enol}}{t \times I_a}$$

where t is the time of irradiation in seconds, and I_a the amount of light absorbed by the biacetyl solution in quanta/sec calculated from the actinometry. The phototube is necessary to determine what fraction of light incident upon the reaction cell is absorbed by the biacetyl solution.

Results

The results obtained by the trial set of students varied from $Q_{enol} = 0.08$ – 0.12 , in reasonable agreement with the published value. The variation in results provides a basis for the discussion of possible errors, the main ones being due to fluctuation in lamp intensity, inefficient removal of oxygen from the biacetyl, and errors due to stray light which affected the extremely light sensitive actinometer solution. The fact that the enolization is quenched by oxygen can be used to initiate a consideration of the properties of the triplet state, which is responsible for this particular reaction.

Acknowledgment

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